



"While climate change is a global driver leading to extreme weather events, human activities act as local amplifiers"

— RAGHUNATH MURTUGUDDE, *Professor Emeritus, University of Maryland*



"The mindset that nothing will happen is the biggest socio-behavioural problem in the country. The culture of safety must be built on a war footing among local bodies and citizens"

— SANJAY SRIVASTAVA, *Chairman, CROPC*



"We should have evacuation strategies for all extreme weather events. Currently, we have only one, for cyclones along the coasts. There is none for floods. All we do is put up tents in safe areas"

— K.J. RAMESH, *Former Director General, India Meterological Department*



"There is a pressing need to consider the consequences of a 2°C [rise in temperature] because, at our current pace, India will not be able to adapt fast enough"

— ABINASH MOHANTY, *Global Sector Head, Climate Change & Sustainability, IPE Global*

that in the past took thousand of years to just a few decades now, says Roxy Mathew Koll, climate scientist at the Indian Institute of Tropical Meteorology, Pune. In fact, the science behind climate change is sobering. For every 1°C rise in global average temperature, the atmosphere can hold 7 per cent more moisture. The oceans, heating faster than ever, evaporate more water, fuelling cyclones and cloudbursts. These create storms that are wetter and more destructive. Hotter oceans see the Earth's frozen poles melt at an even faster rate, driving the rise in sea levels.

WHY THE CLIMATE CHAOS

Averages tell only part of the story. As important is where and when the warming happens. West Asia and the Mediterranean, for example, are heating at a faster pace as dry regions warm faster. This reduces the atmospheric pressure near the surface, creating a strong pressure difference between West Asia and the Arabian Sea, pushing northward the powerful wind systems called low-level jet stream. That shift creates scorching pre-monsoon heat in northwest India and the Himalayan foothills. In the monsoon months, it sprays those same regions with excess moisture. The result is a cruel double whammy of lethal heat waves and catastrophic floods hitting the same areas in back-to-back seasons. "Both seasons are now rendered more extreme due to these perfect conditions over the arid and semi-arid northwest and vulnerable Himalayan foothills," says climate expert Raghunath Murtugudde, currently emeritus professor at the University of Maryland and visiting professor at IIT Kanpur. At the same time, rising temperatures are weakening the Atlantic Meridional Overturning Circulation—a major system of ocean currents—by adding fresh water from melting polar ice, which dilutes seawater and slows the northward flow of warm water. This disruption can shift tropical rain belts and weaken the Indian monsoon, intensifying heat waves and droughts across the subcontinent.

While climate change is a global driver leading to extreme weather events, a country's development activities act as local amplifiers. Urbanisation, infrastructure development and human habitation on mountain slopes exacerbate the impact of rain with more land slips and flash floods. That fallout is visible across India's geography today. In Himachal and Uttarakhand, fragile slopes have been weakened further by highways, tunnels and dams that are collapsing under extreme rainfall, burying entire villages (see accompanying story, *Flood of Apathy*). In Dharali, Uttarakhand, this August, a Himalayan stream turned into a raging torrent, wiping out shops and homes built right on the riverbed. "First, the riverbed was taken over by construction, and now the river has taken it back," says Manshi Asher, an environmental justice activist and researcher and cofounder of Himdhara Collective. In Rajasthan, Punjab and Haryana, the paradox is starker—searing heat waves